

Improving the Distance-Extension Methods for Reliable Wireless Communication in IoT Networks with UAV

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Abstract—The IoT and UAV are the most developed information and communication technologies. The UAV operates in a limited geographical area due to physical and communication channel resource limitations. Thus, a technical task exists to invent and enable methods for increasing UAV operating distance. This paper outlines the distance increase methods for reliable wireless communication in IoT networks with UAVs. The paper describes a practical case of increasing communication distance while satisfying reliable wireless communication requirements in IoT networks with UAVs.

Keywords—IoT, UAV, communication distance

I. INTRODUCTION

A. Definitions

1) IoT networks

IoT is a network of physical devices equipped with sensors, electronics, software, and communication technologies that interact with one another or the external environment. It is designed and built to disrupt economic and social processes and eliminate the need for human participation, which leads to increased efficiency and reduced human workloads [1].

2) UAV

World trends in modern information and telecommunication technology development are expanded in [2-3]. One of the main trends is communication for unmanned aerial vehicles (UAVs). UAVs, also commonly known as drones, or remotely piloted aircrafts, with their high mobility and low cost have found a wide range of applications during the past few decades [4].

3) UAV networks

The next technological evolution of UAVs is the UAV operating in the network rather than in the PtP mode of Operator-UAV. Frequently, such applications are named FANET (flying ad-hoc network) or IoT network with UAV [5].

4) Reliable wireless communication

Reliable communication ensures that the data transmission process satisfies the data quality criteria: accuracy, completeness, relevance, and timeliness. Additionally, reliable communication should satisfy three fundamental bases of information security, the CIA triad: confidentiality, integrity, and availability. Wireless channels for data transmission are one of the communication channel media used for communication with or between UAVs.

B. Problem Statement

Sometimes, UAVs don't satisfy communication distance expectations or other operational requirements (e.g., flight time length, carrying capacity), which requires applying methods and techniques to extend the communication distance with UAVs even further.

C. Task Statement

Research the existing methods of communication with UAVs and increase communication distance by satisfying reliable wireless communication requirements in IoT networks with UAVs.

D. Resources Allocation for Reliable Communication

Communication channel resources for information transmission [6]:

- Energy: limited radiation power, P , W.
- Transmission frequency and frequency band: limited radiation band, ΔF , Hz;
- Time: source symbol rate, V_s , bps.

Essential parameters and techniques for the communication system design and operation:

- Data bit error rate required, e.g., $BER < 10^{-6}$.
- Transmission mode, e.g. OFDM, with subcarrier number N_{OFDM} .
- Multiposition keying (MPK), e.g., QPSK, QAM-16/64: $M = 2^m$.
- Errors correction method, e.g., BCH, LDPC, with parameters n, r_k, t .

Any wireless transmission system consumes limited physical resources of the communication channel: energy, frequency, and time. All the techniques of transmission mode, e.g., multiposition keying and error correction, etc. – use and distribute channel resources in a specific manner and impact the final communication range, data transmission quality, data rate, and others.

II. DISTANCE INCREASE METHODS OVERVIEW FOR COMMUNICATION WITH UAV

Many methods and combinations exist for designing wireless communication systems for UAVs and satisfying distance expectations and requirements.

Such methods can be classified into two major following categories:

1. Communication channel resources distribution.
2. Computational methods in UAV for extending the UAV operating zone beyond the communication link capabilities.

Some of these methods are expanded below.

A. Channel Resources Distribution Techniques to Increase Reliable Communication Range

Let's review some of the communication channel resource distribution techniques to achieve longer communication distances:

1. **Directional antennas with optimal positioning and beamforming** focus signal in a specific direction, increasing range and reducing interference [7].
2. **Signal amplifiers and power control.** They boost the strength of the transmitted signal, extending the communication range; they adjust the transmission power dynamically to optimize range and reduce interference [8, 9].
3. **Adaptive modulation and channel coding for error correction.** Dynamically adjusts modulation and coding schemes based on channel conditions to maintain reliable communication [6, 10, 11].
4. **MIMO (Multiple Input, Multiple Output).** It uses multiple antennas to increase signal strength and data rates, improving range [12].
5. **Frequency-hopping Spread Spectrum (FHSS)** increases *resistance to interference and improves range* by rapidly switching frequencies during transmission [13, 14].
6. **Orthogonal Frequency Division Multiplexing (OFDM)** utilises multiple sub-carriers at different frequencies to improve *resistance to interference and increase range* [6, 15].
7. **Decrease the signal interference conditions: fly higher** to reduce impact from interference or *avoid objects on the communication path*.
8. **Cognitive Radio.** Adapts the communication parameters based on the environment to enhance range and reliability [16, 17].
9. **Low-power wide-area Networks (LPWANs)** use low-power and narrowband techniques to enable long-range communication with minimal power consumption [18].
10. **The relay approach** uses intermediate relay nodes to extend the coverage area and improve signal quality [19, 20].

All these techniques are designed to enable reliable wireless communication with UAVs. However, the operating distance for UAVs sometimes requires even longer distances beyond communication channel capability. In such cases, computational methods and techniques can be applied to extend the UAV operational distance based on navigation systems, calculation algorithms, and their combinations. Some of such methods are described in the next section.

B. Computational Methods for Extending the UAV Operating Zone beyond the Communication Link Capabilities

The use case is the UAV operation when the communication link stops working due to a long distance from

the operator or due to a suppressed communication control channel signal, thus making the UAV an autonomous UAV without any operator help or instructions. In such scenarios, the following computational techniques might be a solution:

- GPS navigation and pursuing the flight goal based on the implemented algorithm to pursue the target or return to the UAV operator if communication was lost with the UAV (automatic return-to-home, RTH);
- Purely relying on the algorithm to pursue the target or return back to the UAV operator if the communication was lost with the UAV and if the GPS signal is suppressed.

While these methods are possible and usually pursue the long-distance carriage delivery goal by UAV, they are not being reviewed extensively in this paper as these techniques are out of the communication link range extension.

Thus, the **mathematical function of the task statement** for increasing communication distance while satisfying reliable wireless communication requirements in IoT networks with UAV is the following (1):

$$\max_{\{\Delta F, P, V_S, Aircraft\}} L \left| \begin{array}{l} BER_{req} \\ OFDM \\ MPK \\ ErrCorr, \\ MIMO \\ FHSS \\ Relay \\ Compute \end{array} \right. \quad (1)$$

where L – is the communication distance with UAV, the target function to maximise, with the limited resources and parameters: F , P , V (or T , time), and *Aircraft* – the UAV capabilities to carry the transmitter away from the operator which is also limited; with parameters that may vary in the allocated resources and supply different BER and data rate: BER_{req} – required Ber, *OFDM*, *MPK*, *ErrCorr* – error correction method, e.g., LDPC; *MIMO*; *FHSS*; *Relay* – using intermediate nodes for signal amplification or handover and thus extending the communication range; and *Compute* – algorithms that allow extending communication and operation range when communication link stops working permanently or temporarily.

Overall, it is worth mentioning that a lower frequency is preferable for long-range communication, as the range will be higher under the same all other conditions (power, bandwidth, etc.). Friis transmission equation [5], which relates the transmitted and the received powers (2):

$$P_{receive} = P_{transmit} G_{transmit} G_{receive} \left(\frac{\lambda}{4\pi L} \right)^2, \quad (2)$$

where L is the separation distance between a transmitter and a receiver in meters, $G_{receive}$ is the gain of the receiving antenna, and λ is the wavelength in meters.

Thus, let's compare some existing UAV models on the market and a practical case of using UAVs with the communication channel operating distance extension.

C. Comparative Overview of UAV and Communication Control Distance

Here is a comparative overview of various UAV models from varied brands [21], highlighting their maximum control communication ranges and flight times (Table 1).

TABLE I. VARIOUS UAV MODELS: MAX CONTROL RANGE AND FLIGHT TIME

Product	Max Control Range	Max Flight Time
DJI Matrice 350 RTK	20 km	55 minutes
DJI Matrice 30	7 km	41 minutes
DJI Mavic 3 Pro	15 km	43 minutes
DJI Air 3	20 km	46 minutes
DJI Mini 4 Pro	20 km	45 minutes
DJI Avata 2	13 km	23 minutes
DJI Phantom 4 RTK	7 km	30 minutes
DJI Agras T30	5 km	20.5 minutes
DJI Tello	0.1 km	13 minutes
T-DRONES VA23	30 km	240 minutes
Parrot ANAFI Ai	22 km	32 minutes
Parrot Bebop 2	2 km	25 minutes
Autel EVO II Pro V3	15 km	40 minutes
Autel EVO Max	20 km	42 minutes
Autel Dragonfish Pro	30 km	158 minutes
Holy Stone HS720	0.9 km	26 minutes
Holy Stone HS600	3 km	56 minutes

As Table 1 shows, there are significant variations in both maximum control ranges and flight times. For example, some models propose up to 30 km and up to 240 minutes. These features make them highly suitable for professional and commercial use, where extensive operational capacity and broad area coverage are crucial. On the other hand, some models propose a 0.1 km range and 13 minutes of flight time. Thus, these models are more suitable for local tasks.

III. PRACTICAL CASE OF USING UAV AND SIGNAL DISTANCE EXTENSION

UAVs are increasingly used in civil and military applications. It is worth highlighting that UAVs played a key role in the war caused by the Russian Federation in Ukraine. They are used for a wide range of applications, such as video surveillance, explosive materials delivery, medical help supply, and many others. Therefore, UAVs play a vital role in modern technological wars.

The analysis below expands the use case of the UAV DJI Mavic 3 and the communication range increase via the Aliantech Duo 3 antenna signal booster range extender on the front line in the war of the Russian Federation in Ukraine. Let's review the technical characteristics of this UAV solution with an extended range.

A. UAV: DJI Mavic 3

The UAV DJI Mavic 3 is shown in Fig. 1 [22].

First, it is worth mentioning that the operating distance of UAVs heavily depends on the aircraft's design and physical capabilities to operate for a long time and distance.

For this particular model of UAV DJI Mavic 3, as stated in [22], some of the key operational, communication and data transmission characteristics are the following:

- Takeoff weight Mavic 3: 958 g.



Fig. 1. UAV DJI Mavic 3.

- Max flight distance: 28 km; flight time: 43 minutes.
- Global Navigation: GPS, Galileo and BeiDou.
- Video Transmission System: O3+.
- Operating Frequency: 2.4-2.4835, 5.725-5.85 GHz.
- Transmitter Power (EIRP): 2.4 GHz: < 33 dBm (FCC), < 20 dBm (CE/SRRC/MIC), 5.8 GHz: < 33 dBm (FCC), < 30 dBm (SRRC), < 14 dBm (CE).
- Max Transmission Distance (use for reference purposes only - no guarantee for actual transmission distance):
 - Unobstructed, free of interference: FCC: 15 km; CE: 8 km; SRRC: 8 km; MIC: 8 km.
 - Unobstructed, with interference: Strong Interference: Urban landscape, approx. 1.5-3 km; Medium Interference: Suburban landscape, approx. 3-9 km; Low Interference: Suburb/seaside, approx. 9-15 km.
 - Obstructed, with interference: Low Interference and Obstructed by Buildings: approx. 0-0.5 km; Low Interference and Obstructed by Trees: approx. 0.5-3 km
- Max Download Speed: O3+: up to 15 MB/s; Wi-Fi 6: 80 MB/s.
- Antenna: 4 antennas, 2T4R (MIMO).

The generic communication link scheme for work with the UAV DJI Mavic 3 is shown in Fig. 2.

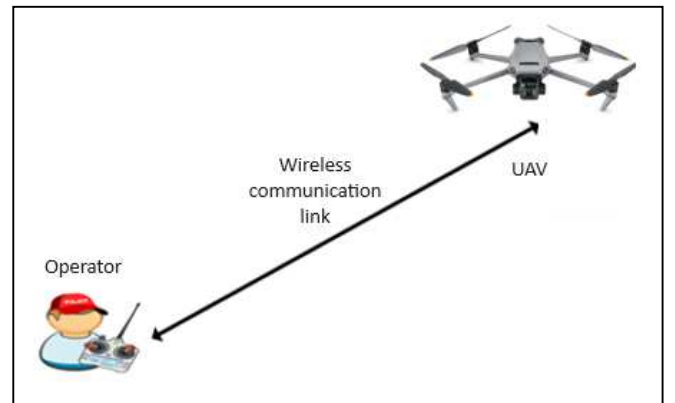


Fig. 2. Aliantech Duo 3 antenna signal booster range extender, cable and Controller for UAV DJI Mavic 3.

However, as practice has shown, these characteristics above satisfy the military operation on the front line, usually not fully due to limited operation range, this UAV is designed for civil applications mainly and not for military use, but it is still exceptionally well operated in the field conditions for military operations. Thus, modernization is required for this solution to achieve better technical and operational characteristics, such as communication for a longer distance when returning the UAV to the operator. Let's review the signal distance increase by using the signal booster.

B. Signal Distance Increase for UAV DJI Mavic 3 via Alientech Duo 3 Antenna Signal booster Range Extender

One method for increasing distance to achieve reliable wireless communication with UAVs is the Alientech Duo 3 antenna signal booster range extender [23] for the DJI Mavic 3 UAV.

MIMO technology with dual transmission and reception capabilities improves signal strength and better overall performance, even in weak signal areas.

The DUO 3 Signal Booster (Fig. 3, Fig. 4) operates simultaneously in the 2.4G, 5.2G, and 5.8G frequency bands. The device is compliant with IEEE 802.11a/b/g/n/ac standards. The DUO 3 Signal Booster supports *spread spectrum, multi-carrier modulation, time-division duplexing, automatic frequency switching, channel selection, and multi-channel support*, enhancing *signal strength and quality* while minimizing interference. The device can extend the transmission distance and improve the *remote control range*. It is FCC and CE-compliant, ensuring it meets the regulatory requirements for transmission power and signal quality.



Fig. 3. Alientech Duo 3 antenna signal booster range extender, cable and Controller for UAV DJI Mavic 3 .

Some of the key technical specifications:

- Antenna Gain: 2.4G>13 dBi±1 / 5G>15 dBi±1
- 2.4G Receive Gain: 12dB±1
- Receive Gain of 5.2G and 5.8G: 15dB±1
- Transmit Gain: 15 dB±1
- Input power range: 10–25 dBm
- EVM: 3%@28 dBm 802.11 ac 6.93Gbps MIMO OFDM 64QAM BW 80 MHz



Fig. 4. Alientech Duo 3 antenna signal booster range extender for UAV DJI Mavic 3 in the field conditions.

As a result, due to using a signal distance increase via Alientech Duo 3 antenna signal booster range extender for UAV DJI Mavic 3, - the signal range gets a boost equivalent to 5-10 times with the stock antenna, which is a practical case of communication range increase. This solution has been tested and operates successfully in the field conditions on the front line of the russian federation's war in Ukraine.

Thus, **the aim of the work is achieved**, i.e., research on the existing methods of communication with UAV has been done, and increasing communication distance while satisfying reliable wireless communication requirements in IoT networks with UAV is shown in practice.

The described use case covers the PtP UAV-to-operator communication scenario. However, UAVs can be used not only in PtP mode with an operator but also in a network, which is expanded in the next section.

C. Extending UAV application to IoT Networks with UAV

Multiple UAVs connected in a wireless ad hoc manner form a flying ad hoc network (FANET) or IoT network with UAVs, offering multi-hop connectivity [5]. UAVs can be equipped with different networking technologies for communication between multiple UAVs and other systems, such as ground control stations (GCSs) and wireless sensor networks (WSNs). Specifically, in a FANET, two main types of communication paradigms can be highlighted:

- UAV-to-UAV (3U) communications, where UAVs communicate with each other via line of sight (LoS) propagation links in a multihop manner to extend the communications range;
- UAV-to-GCS communications (3GCS), where a UAV communicates directly with the GCS.

Mixing these two communication paradigms is the basis of hybrid approaches, which represent a combination of 3U and 3GCS communications, and where the UAV can send its data directly to the GCS in a one-hop or multi-hop fashion via the different UAVs in the mission area. UAVs can then function as relay nodes for data forwarding in a FANET.

It is worth mentioning that such scenarios of IoT Networks with UAVs are much more complicated in implementation and operation than the scenario of PtP UAV to operator communication analyzed in this paper. They also heavily rely on the communication algorithm design in addition to the communication channel resources and capabilities.

IV. CONCLUSION

1. The challenge of UAVs not always meeting communication distance expectations or other operational requirements is significant. This necessitates applying methods and techniques for extending the communication distance with UAVs, making our research even more crucial.
2. Communication channel resource allocation for reliable communication has been outlined.
3. This paper meticulously reviews the existing methods of communication with UAVs and the increase in communication distance required to satisfy reliable wireless communication requirements in IoT networks with UAVs. It also provides a comprehensive overview of the top ten most frequently used methods for signal range extension.
4. It has been emphasized that communication range can be increased following two main paradigms: via the distribution of communication channel resources or additional computational methods in UAV to extend the UAV operating zone beyond the communication link capabilities. The current paper focused more on the paradigm of communication channel resource distribution.
5. The mathematical function of the task statement for increasing communication distance while satisfying reliable wireless communication requirements in IoT networks with UAVs has been defined.
6. A comparative overview of various UAV models from varied commercial brands has been performed, highlighting their maximum control communication ranges and flight times.
7. A practical case has been described in detail with technical characteristics of the UAV DJI Mavic 3 use with communication range increase via the Alientech Duo 3 antenna signal booster range extender. This solution has been tested and operates successfully in the field conditions on the front line of the Russian Federation's war in Ukraine. As a result, the signal range gets a boost of 5-10 times with the stock antenna, giving the power to reach new heights and explore uncharted territories. Thus, the aim of the work is achieved, i.e., research on the existing methods of communication with UAV has been done, and increased communication distance with satisfying reliable wireless communication requirements in IoT networks with UAV has been shown in practice.
8. Scenarios of IoT Networks with UAVs are much more complicated in implementation and operation compared to the scenario of PtP UAV to Operator communication. They also heavily rely on the design of the communication algorithms and the communication channel resources and capabilities.

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